

Xingjian Diao

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 [LinkedIn](#) |  [Github](#) |  [Google Scholar](#)

RESEARCH INTERESTS

My research focuses on **multimodal learning** for video, audio, and language understanding. I have developed methods for **multimodal reasoning**, **efficient multimodal learning**, and **multimodal large language model post-training**, aiming to build scalable and generalizable multimodal models that advance multimodal QA, video understanding, and VLM-based GUI agents across complex real-world scenarios and dynamic environments. My GitHub repositories on multimodal large language models (MLLMs) have received 1.5k+ Stars ★.

EDUCATION

- **Dartmouth College** Sep 2022 - Mar 2027 (Expected)
Ph.D. candidate in Computer Science Hanover, NH, USA
 - **Advisor:** [Prof. Soroush Vosoughi](#) and [Prof. Jiang Gui](#)
- **Northwestern University** Sep 2020 - Dec 2021
Master of Science, Computer Science Evanston, IL, USA
 - **Advisor:** [Prof. Nabil Alshurafa](#)
- **University of Pittsburgh** Aug 2016 - Apr 2020
Bachelor of Science, Computer Science Pittsburgh, PA, USA

INTERNSHIP

- **Amazon** Jun 2026 - Sep 2026
Applied Scientist Intern @ RBKS AI Sunnyvale, CA, USA

VISTA-GUI: Verifiable Interaction State-Transition Alignment for Mobile GUI Agents
Building VISTA-GUI, a mobile GUI agent framework that converts interaction trajectories into structured task states, expected action effects, observable evidence, and recovery policies. It augments expert trajectories with counterfactual failures, uses multi-level verifiers to produce process rewards, and trains the agent through transition supervision, preference optimization, and online RL, enabling reliable progress verification and error recovery in long-horizon mobile tasks.
💡 [GUI Agents](#) 💡 [VLM](#) 💡 [Reinforcement Learning](#)
- **Samsung Research America** Mar 2026 - Jun 2026
NLP Research Intern @ Language Intelligence Team, AI Center Mountain View, CA, USA

Doc-to-Atom: Learning to Compile and Compose Memory Atoms
[Pdf](#) | Built Doc-to-Atom, a compositional parametric-memory framework that internalizes a document into a base LLM by decomposing it into semantically typed knowledge atoms, each compiled into a micro-LoRA adapter and a retrieval key via a hypernetwork; a two-stage router then assembles only the query-relevant atoms into a query-specific adapter at inference. Trained end-to-end with a multi-objective distillation curriculum, it outperforms document-internalization (document-to-LoRA) baselines across six diverse QA benchmarks while using up to 85% less compile-time GPU memory.
💡 [Parametric Memory](#) 💡 [Hypernetwork](#) 💡 [Context Distillation](#)
- **Amazon** June 2025 - Sept 2025
Applied Scientist Intern @ Last Mile (Full-Time Return Inclined) Santa Cruz, CA, USA

High-Frequency Video-to-IMU Synthesis via Physics-Guided Simulation and Hybrid U-Net Refinement
[Pdf](#) | Proposed PrimeIMU, a physics-guided video-to-IMU generation framework that fuses low-frequency kinematic cues from 3D video poses with physics-inspired simulated inertial initialization through a hybrid U-Net refinement module, effectively bridging the anatomical-inertial gap to produce high-fidelity, sensor-faithful IMU signals that generalize across activities, devices, and datasets, enabling synthetic-only training, cross-domain adaptation, and scalable deployment of wearable sensing models.
💡 [Computer Vision](#) 💡 [Ubiquitous Computing](#)

IMU2Reason: A Multimodal LLM for Safety-Aware Activity Understanding
[Pdf](#) | Proposed MHIA-LM, a multimodal LLM trained on fused video and IMU representations, where spatiotemporal video embeddings from InternVideo2 and IMU motion features encoded by a PatchTST-based encoder are integrated through a multi-layer Gated Hierarchical Interaction module and mapped into the LLM token embedding space for instruction tuning, enabling fine-grained activity recognition and safety reasoning.
💡 [MLLM](#) 💡 [Multimodal Reasoning](#)

[Preprint 2026] **Doc-to-Atom: Learning to Compile and Compose Memory Atoms**

Xingjian Diao, Wenbo Li, Yashas Malur Saidutta, Avinash Amballa, Lazar Valkov, Srinivas Chappidi

[Pdf](#) | Proposed Doc-to-Atom, a compositional parametric-memory framework that decomposes each document into semantically typed knowledge atoms, compiles them via a hypernetwork into per-atom micro-LoRA adapters with retrieval keys, and trains a two-stage router to assemble only the query-relevant atoms into a query-specific adapter, outperforming document-to-LoRA baselines across six QA benchmarks at up to 85% less GPU memory.

💡 [Parametric Memory](#) 💡 [Hypernetwork](#) 💡 [Context Distillation](#)

[ACL 2026] **Addressing Overthinking in Large Vision-Language Models via Gated Perception-Reasoning Optimization**

BMDs Travel Award, Dartmouth College

Xingjian Diao, Zheyuan Liu, Chunhui Zhang, Weiyi Wu, Keyi Kong, Lin Shi, Kaize Ding, Soroush Vosoughi, Jiang Gui

[Pdf](#) | Introduced Gated Perception-Reasoning Optimization (GPRO), a token-level adaptive computation framework that leverages large-scale perception-reasoning failure attribution ($\approx 790K$ samples) to train a meta-reasoning controller via multi-objective reinforcement learning, dynamically routing between fast execution, visual re-perception, and slow reasoning to mitigate overthinking while improving both accuracy and efficiency in vision-language models.

💡 [Large Vision-Language Model](#) 💡 [Overthinking Mitigation](#) 💡 [Reinforcement Learning](#)

[EMNLP 2025] **SoundMind: RL-Incentivized Logic Reasoning for Audio-Language Models**

Oral Presentation Award (top 4.35%), 30th EMNLP

Xingjian Diao, Chunhui Zhang, Keyi Kong, Weiyi Wu, Chiyu Ma, Zhongyu Ouyang, Peijun Qing, Soroush Vosoughi, Jiang Gui

[Pdf](#) | [Github Code](#); ★ **Starred 1k+** | [Dataset](#) | Introduced the Audio Logical Reasoning (ALR) task containing 6,446 text-audio CoT-annotated samples to enable complex reasoning over spoken content, and proposed SoundMind, a rule-based reinforcement learning algorithm that enhances deep cross-modal reasoning in audio-language models.

💡 [Large Audio-Language Model](#) 💡 [Logical Reasoning](#) 💡 [Reinforcement Learning](#)

[EMNLP 2025] **ProtoVQA: An Adaptable Prototypical Framework for Explainable Fine-Grained Visual Question Answering**

Oral Presentation Award (top 4.35%), 30th EMNLP

Xingjian Diao*, Weiyi Wu*, Keyi Kong, Peijun Qing, Ming Cheng, Soroush Vosoughi, Jiang Gui

[Pdf](#) | Proposed ProtoVQA, an adaptable prototypical VQA framework that learns question-aware prototypes and uses spatially-constrained greedy matching to ground answers in semantically coherent image regions, unifying answering and grounding via a shared backbone and introducing the VLAS metric to quantify visual-linguistic alignment.

💡 [Multimodal QA](#) 💡 [Interpretability](#)

[NAACL 2025] **Temporal Working Memory: Query-Guided Temporal Segment Refinement for Enhanced Multimodal Understanding**

Guarini Graduate Student Travel Award, Dartmouth College

Xingjian Diao*, Chunhui Zhang*, Weiyi Wu, Zhongyu Ouyang, Peijun Qing, Ming Cheng, Soroush Vosoughi, Jiang Gui

[Pdf](#) | [Github Code](#); ★ **Starred 300+** | Proposed Temporal Working Memory, a plug-and-play query-guided segment refinement module that maintains dynamic temporal memory to effectively preserve task-relevant video-audio segment and enhance long-range temporal reasoning, achieving consistent performance gains when integrated into nine recent state-of-the-art multimodal large language models (MLLMs) across AVQA, video captioning, and retrieval tasks.

💡 [MLLM](#) 💡 [Video Understanding](#)

[EMNLP 2024] **Learning Musical Representations for Music Performance Question Answering**

BMDs Travel Award, Dartmouth College

Xingjian Diao, Chunhui Zhang, Tingxuan Wu, Ming Cheng, Zhongyu Ouyang, Weiyi Wu, Jiang Gui

[Pdf](#) | Proposed a specialized framework for audio-visual modeling in music understanding, addressing underexplored multimodal interactions, distinctive musical characteristics, and temporal alignment, and introduced annotated rhythmic and source features, with the framework achieving state-of-the-art on Music-AVQA 1.0 and 2.0.

💡 [Multimodal QA](#) 💡 [Representation Learning](#)

[ACL 2025] **Learning Sparsity for Effective and Efficient Music Performance Question Answering**

Xingjian Diao, Tianzhen Yang, Chunhui Zhang, Weiyi Wu, Ming Cheng, Jiang Gui

[Pdf](#) | Proposed Sparsify, a sparse learning framework for Music Audio-Visual Question Answering that integrates Sparse Masking, Adaptive Sparse Merging, and Sparse Subset Selection to reduce multimodal redundancy, highlight task-critical tokens, and accelerate training convergence, achieving efficiency and accuracy gains across Music-AVQA benchmarks.

💡 [Multimodal QA](#) 💡 [Sparsity Learning](#)

[ACL 2026] **Music Performance Audio-Visual Question Answering Requires Specialized Multimodal Designs**

Wenhao You*, Xingjian Diao*, Wenjun Huang, Chunhui Zhang, Keyi Kong, Weiyi Wu, Chiyu Ma, Zhongyu Ouyang, Tingxuan Wu, Ming Cheng, Soroush Vosoughi, Jiang Gui

[Pdf](#) | Providing a systematic analysis of how specialized multimodal architectures with spatio-temporal modeling enable reliable reasoning over musical performances.

💡 [Multimodal QA](#) 💡 [Video Understanding](#)

[WACV 2025] FT2TF: First-Person Statement Text-To-Talking Face Generation

Xingjian Diao, Ming Cheng, Wayner Barrios, SouYoung Jin

[Pdf](#) | Proposed and developed a one-stage end-to-end text-to-talking face generation pipeline driven by first-person statement text, requiring only visual and textual inputs during inference. Experiments on LRS2 and LRS3 demonstrate state-of-the-art performance, showing its ability to generate realistic talking faces effectively from text inputs.

💡 AIGC 💡 Multimodal Alignment

SELECTED COLLABORATIVE PUBLICATIONS

[Preprint 2026] Linear Attention Scales Better with Shared Latent Memory Projection

Chunhui Zhang, Xingjian Diao, Xiangchi Yuan, Chiyu Ma, Yaning Jia, Zhongyu Ouyang, Soroush Vosoughi
Proposed Mixture-of-Latent-Memories (MoLM), a scalable multi-memory linear-attention architecture that compresses each token into a shared latent representation and decodes it into memory-specific key-value states through lightweight per-memory projections, decoupling memory count from projection cost and enabling higher throughput, lower GPU memory usage, and stronger long-context performance than Mixture-of-Memories baselines.

💡 Linear Attention 💡 Mixture-of-Memories

[CVPR 2026] Learning Spatial-Preserving Hierarchical Representations for Digital Pathology

Weiye Wu, Xingjian Diao, Chunhui Zhang, Chongyang Gao, Xinwen Xu, Siting Li, Jiang Gui

[Pdf](#) | Introduced Sparse Pyramid Attention Networks (SPAN) for gigapixel whole-slide pathology, combining spatial-adaptive condensation with context-aware refinement to preserve spatial structure and enable efficient multi-scale tumor detection, classification, and segmentation.

💡 Digital Pathology 💡 Whole Slide Image Analysis

[ICML 2026] HiST: A Hierarchical Sparse Transformer for Cross-Modal Spatial Transcriptomics Modeling

Weiye Wu, Xinwen Xu, Xingjian Diao, Siting Li, Zhi Wei, Alma Andersson, Jiang Gui

[Pdf](#) | Proposed HiST, a hierarchical sparse transformer for H&E-to-ST inference that models measured spots as a lattice-indexed sparse field to capture multiscale tissue context efficiently, improving predictive performance while substantially reducing inference time and memory across multi-organ spatial transcriptomics benchmarks.

💡 Spatial Transcriptomics

[ACL 2026] What Makes a Good Curriculum? Disentangling the Effects of Data Ordering on LLM Mathematical Reasoning

Yaning Jia, Chunhui Zhang, Xingjian Diao, Xiangchi Yuan, Zhongyu Ouyang, Chiyu Ma, Soroush Vosoughi

[Pdf](#) | Proposed a unified offline curriculum learning framework that systematically disentangles five curriculum dimensions and isolates the causal impact of data ordering to provide principled guidance on curriculum design.

💡 LLM 💡 Curriculum Learning

[EACL 2026] Tailoring Memory Granularity for Multi-Hop Reasoning over Long Contexts

Peijun Qing, Xingjian Diao, Chiyu Ma, Saeed Hassanpour, Soroush Vosoughi

[Pdf](#) | Proposed Tailoring Memory Granularity, a reward-guided framework that adaptively composes hybrid memory across multiple granularities, including chunks, triples, atomic facts, and summaries, to enhance large language models in long-context multi-hop reasoning through dynamic query-specific memory selection.

💡 LLM 💡 Hybrid Memory 💡 Multi-hop Reasoning

[TMLR 2026] The Pitfalls of Text Degeneration when Aligning LLMs through Model Merge

Peijun Qing, Hefan Zhang, Lei Hsiung, Haiquan Lu, Xingjian Diao, Chiyu Ma, Saeed Hassanpour, Soroush Vosoughi

[Pdf](#) | Systematically investigated how alignment-oriented model merging in decoder-based LLMs can induce localized text degeneration, including repetition, unintended code switching, and nonsensical outputs, even when standard reward and performance metrics improve, through a large scale analysis of extrapolation and interpolation merges across multiple models and merging methods.

💡 Model Merge

[WACV 2026] Exploiting Label-Independent Regularization from Spatial Patterns for Whole Slide Image Analysis

Weiye Wu, Xinwen Xu, Chongyang Gao, Xingjian Diao, Siting Li, Jiang Gui

[Pdf](#) | Proposed SRMIL for whole-slide image analysis, combining label-guided classification with label-independent spatial reconstruction to exploit spatial patterns as regularization, improve feature learning, and enhance robust weakly supervised classification.

💡 Digital Pathology

[AAACL 2025] Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge

Oral Presentation Award (top 3.30%), 14th IJCNLP-AAACL

Lin Shi, Chiyu Ma, Wenhua Liang, Xingjian Diao, Weicheng Ma, Soroush Vosoughi

[Pdf](#) | Proposed a systematic position-bias evaluation framework for LLM-as-a-Judge that integrates Repetition Stability, Position Consistency, and Preference Fairness to measure bias robustness, expose primacy-recency behaviors under prompt permutations, and extend analysis from pairwise to list-wise comparisons on MTBench and DevBench.

💡 LLM-as-a-Judge

- [EMNLP 2025] Knowing More, Acting Better: Hierarchical Representation for Embodied Decision-Making**
 Chunhui Zhang, Zhongyu Ouyang, **Xingjian Diao**, Zheyuan Liu, Soroush Vosoughi
[Pdf](#) | Proposed a hierarchical action probing framework that aggregates layer-wise MLLM representations to enhance spatial grounding, contextual integration, and abstract reasoning for embodied decision-making, achieving substantial gains across language-guided rearrangement tasks.
 💎 [MLLM](#) 💎 [Vision–Language–Action](#)
- [EMNLP 2025] Assessing and Mitigating Medical Knowledge Drift and Conflicts in Large Language Models**
 Weiyi Wu, Xinwen Xu, Chongyang Gao, **Xingjian Diao**, Siting Li, Lucas A Salas, Jiang Gui
[Pdf](#) | Proposed ConflictMedQA, a guideline-grounded benchmark pairing up-to-date and outdated clinical recommendations, along with ECDA/IKCR metrics to evaluate medical knowledge drift and assess RAG, DPO, and RoD for resolving temporal conflicts in LLMs.
 💎 [Medical LLM](#) 💎 [Knowledge Drift](#) 💎 [RAG](#)
- [Preprint 2025] On The Design Choices of Next Level LLMs**
 Yijun Tian, **Xingjian Diao**, Ming Cheng, Chunhui Zhang, Jiang Gui, Soroush Vosoughi, Xiangliang Zhang, Nitesh V. Chawla, Shichao Pei
[Pdf](#) | Provided a comprehensive analysis of current LLM design choices across model architecture, attention mechanisms, post-training strategies, optimization techniques, and data selection, identifying key trends and proposing future research directions for next-generation large language models.
 💎 [LLM](#) 💎 [Post-Training](#) 💎 [Reinforcement Learning](#)
- [EMNLP 2024] AlphaLoRA: Assigning LoRA Experts Based on Layer Training Quality**
 Peijun Qing, Chongyang Gao, Yefan Zhou, **Xingjian Diao**, Yaoqing Yang, Soroush Vosoughi
[Pdf](#) | Proposed AlphaLoRA, a training-free layer-wise expert allocation strategy for LoRA-MoE that leverages Heavy-Tailed Self-Regularization to quantify layer training quality and assign experts accordingly, reducing redundancy while improving performance across NLP and reasoning benchmarks.
 💎 [LLM](#) 💎 [Mixture-of-Experts](#)
- [EMBC 2024] GluMarker: A Novel Predictive Modeling of Glycemic Control Through Digital Biomarkers**
EMBC NextGen Scholar Award, IEEE 46th EMBC
 Ziyi Zhou*, Ming Cheng*, **Xingjian Diao***, Yanjun Cui, Xiangling Li
[Pdf](#) | Proposed GluMarker, a digital biomarker framework that integrates broader daily factors (meals, insulin doses, and CGM-derived metrics) to predict next-day glycemic control and identify key digital biomarkers that reveal how everyday behaviors shape diabetes management.
 💎 [Digital Biomarkers](#)

TEACHINGS

Graduate Teaching Assistant

- [COSC89/189](#) (Video Understanding), Dartmouth College, Spring 2024
- [COSC74/274](#) (Machine Learning), Dartmouth College, Winter 2024
- [COSC61](#) (Database Systems), Dartmouth College, Summer 2023
- [COSC10](#) (Object Oriented Programming), Dartmouth College, Spring 2023
- [COSC62/162](#) (Applied Cryptography), Dartmouth College, Winter 2023
- [COSC10](#) (Object Oriented Programming), Dartmouth College, Fall 2022

SERVICES

Reviewer / Program Committee Member

Conferences

- ICML (International Conference on Machine Learning) 2026
- NeurIPS (Annual Conference on Neural Information Processing Systems) 2025, 2026
- ICLR (International Conference on Learning Representations) 2025, 2026
- CVPR (Conference on Computer Vision and Pattern Recognition) 2025, 2026
- ICCV (International Conference on Computer Vision) 2025
- ACL (Annual Meeting of the Association for Computational Linguistics) 2025, 2026
- ACL (ACL Industry Track) 2025, 2026
- EMNLP (Empirical Methods in Natural Language Processing) 2025, 2026
- ACM MM (ACM International Conference on Multimedia) 2025, 2026
- ACM MM (Datasets Track) 2025
- TMLR (Transactions on Machine Learning Research) 2025, 2026
- WACV (IEEE/CVF Winter Conference on Applications of Computer Vision) 2025
- AISTATS (International Conference on Artificial Intelligence and Statistics) 2026
- EACL (European Chapter of the Association for Computational Linguistics) 2026
- ICASSP (International Conference on Acoustics, Speech & Signal Processing) 2025, 2026
- IUI (ACM International Conference on Intelligent User Interfaces) 2026
- ISBI (IEEE International Symposium on Biomedical Imaging) 2025, 2026
- BMVC (British Machine Vision Conference) 2026

- IJCNN (International Joint Conference on Neural Networks) 2024, 2025, 2026
- ICME (IEEE International Conference on Multimedia & Expo) 2024, 2025, 2026
- AVSS (IEEE International Conference on Advanced Video and Signal-Based Systems) 2025
- **Journal / Transactions**
- TMLR (Transactions on Machine Learning Research) 2025, 2026
- TAI (IEEE Transactions on Artificial Intelligence) 2026
- **Workshops**
- MINT (NeurIPS Workshop on Foundation Model Interventions) 2024

AWARDS

• Dartmouth Fellowship , Dartmouth College.	2022-Present
• Biomedical Data Science Travel Award , Dartmouth College.	2026
• EMNLP Oral Presentation Award (top 4.35%) , 30 th EMNLP.	2025
• IJCNLP-AAACL Oral Presentation Award (top 3.30%) , 14 th IJCNLP-AAACL.	2025
• Guarini School of Graduate and Advanced Studies Travel Award , Dartmouth College.	2025
• Biomedical Data Science Travel Award , Dartmouth College.	2025
• IEEE EMBC NextGen Scholar Award , IEEE EMBC.	2024